

# Olist E-Commerce Sales Analysis Report

## Executive Summary

This project analyzes the Olist Brazilian E-Commerce dataset to understand sales performance, customer purchasing behavior, and product trends. Using SQL for data preparation and Power BI for visualization, an interactive dashboard was built to help answer key business questions and support data-driven decision-making.

The analysis is based on four core tables:

- Customers
- Orders
- Order Items
- Products

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## Project Objectives

- Analyze overall sales performance.
- Understand customer purchasing behavior.
- Identify top-performing product categories.
- Measure product demand using order data.
- Build an interactive Power BI dashboard for business reporting.

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## Dataset Overview

The project uses the following tables from the Olist dataset:

| Table       | Description                                                                 |
|-------------|-----------------------------------------------------------------------------|
| Customers   | Customer information and unique customer IDs                                |
| Orders      | Order details and purchase information                                      |
| Order Items | Product-level information for each order, including price and freight value |
| Products    | Product details and category information                                    |

# Data Preparation

The dataset was cleaned and transformed before analysis.

The preprocessing steps included:

- Removing duplicate records
  - Handling missing values where necessary
  - Renaming columns for better readability
  - Converting data types
  - Creating relationships between the four tables
  - Validating primary and foreign keys
  - Creating calculated measures in Power BI using DAX
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## Dashboard KPIs

The dashboard includes the following key metrics:

- Total Revenue
  - Total Orders
  - Total Customers
  - Total Items Sold
  - Average Order Value (AOV)
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## Visualizations

The dashboard provides insights through:

- Monthly Sales Trend
  - Revenue by Product Category
  - Top Categories by Items Sold
  - Revenue vs. Items Sold by Category
  - KPI Cards for Revenue, Orders, Customers, Items Sold, and AOV
  - Interactive slicers for better data exploration
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# Key Insights

## 1. Revenue is concentrated in a few product categories.

A small number of categories contribute a significant share of total revenue, indicating where the business should focus inventory and marketing efforts.

## 2. Customer purchasing behavior is uneven.

Many customers make only one purchase, while a smaller group contributes through multiple orders. This suggests an opportunity to improve customer retention.

## 3. Product categories perform differently.

Some categories consistently generate higher sales and revenue, helping identify the strongest-performing product segments.

## 4. Sales show variation over time.

Monthly sales trends reveal periods of higher and lower demand, which can support better inventory planning and promotional strategies.

## 5. Average Order Value helps measure customer spending.

Tracking AOV provides insight into purchasing behavior and can be used to evaluate pricing and promotional effectiveness.

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# Business Recommendations

- Focus marketing efforts on high-performing product categories.
- Introduce customer retention programs to encourage repeat purchases.
- Optimize inventory based on monthly sales patterns.
- Monitor Average Order Value to identify opportunities for upselling and cross-selling.
- Regularly track category performance to improve product strategy.

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# Tools & Technologies

- SQL
- Power BI
- Power Query
- DAX
- Microsoft Excel

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## Skills Demonstrated

- SQL Joins
- Data Cleaning
- Data Modeling
- DAX Measures
- KPI Development
- Business Intelligence
- Data Visualization
- Analytical Thinking
- Dashboard Design

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## Conclusion

This project demonstrates an end-to-end data analytics workflow using SQL and Power BI. By integrating customer, order, order item, and product data, the dashboard delivers meaningful insights into sales performance and customer behavior. The project highlights practical skills in data preparation, modeling, visualization, and business analysis that are directly applicable to real-world data analyst roles.